

Francis Omoniyi Emmanuel

Mathematics Educator & Applied Mathematician

CONTACT INFORMATION

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EDUCATION

University of Alabama at Birmingham, Birmingham, Alabama

Ph.D. in Applied Mathematics

Aug. 2022–Present

- Expected completion: Apr. 2027
- Dissertation: *Mathematical Modeling of Cardiovascular Disease–Mortality Dynamics: A Data-Informed Framework with Physical Activity Effect Modification*

University of Alabama at Birmingham, Birmingham, Alabama

M.S. in Mathematics, Non-Thesis

Aug. 2022–Apr. 2024

University of Ilorin, Ilorin, Nigeria

B.S. in Mathematics, Minor in Statistics

Oct. 2017–Aug. 2021

TEACHING EXPERIENCE

Graduate Teaching Assistant, University of Alabama at Birmingham

Aug.

2022–Present

- Teach undergraduate mathematics courses ranging from developmental algebra through Calculus I.
- Prepare instructional materials, quizzes, examinations, review materials, and other course content.
- Grade student work and provide timely feedback aligned with course expectations and learning objectives.
- Hold office and tutoring hours and provide individualized academic support to students with varied levels of mathematical preparation.
- Use Canvas, ALEKS, Pearson, iClicker, and Poll Everywhere to support course organization, practice, classroom participation, and formative assessment.
- Use student performance on assignments, quizzes, and in-class activities to identify misconceptions and determine when material should be revisited or represented differently.

Upward Bound Mathematics Instructor, University of Alabama at Birmingham

Summer 2026

- Designed and taught Algebra II with Trigonometry and Precalculus for high-school students preparing for college mathematics.
- Developed lesson plans, instructional materials, guided-practice activities, and assessments.
- Adapted explanations and practice to students with varied levels of prior preparation while maintaining appropriate mathematical standards.
- Provided individualized academic support aimed at strengthening mathematical foundations, confidence, and college readiness.

Mathematics Lecturer, Peniel Advanced Pre-Versity, Nigeria

Jan. 2020–Dec. 2021

- Taught Advanced Mathematics I and II through lectures, guided problem solving, discussion, and examination preparation.
- Prepared and graded assessments and held regular office hours.
- Supported students in developing mathematical reasoning, confidence, and problem-solving skills.

COURSES TAUGHT

- MA 125 – Calculus I Fall 2022, Spring 2023, Fall 2026
- MA 107 – Precalculus Algebra/Trigonometry Fall 2025, Spring 2026
- MA 106 – Precalculus Trigonometry Summer 2025

- MA 102 – Intermediate Algebra Spring 2025
- MA 105 – Precalculus Algebra Spring, Summer, Fall 2024
- MA 098 – Basic Algebra Fall 2023
- Algebra II with Trigonometry Summer 2026
- Precalculus Summer 2026
- MAT 111/112 – Advanced Mathematics I and II 2020–2021

TEACHING AWARDS & DEVELOPMENT

- Outstanding Graduate Teaching Assistant**, University of Alabama at Birmingham 2025
- Lean Into Future Learning (LIFT) Teaching Certificate**, Center for Teaching and Learning, University of Alabama at Birmingham 2026
Training in course design, evidence-based pedagogy, student engagement, inclusive teaching, assessment, and reflective practice.
- Best Tutor of the Year**, Faculty of Physical Sciences, University of Ilorin 2021

INSTRUCTIONAL TECHNOLOGY

- **Learning Management and Courseware:** Canvas, ALEKS, Pearson
- **Student Engagement and Assessment:** iClicker, Poll Everywhere
- **Computational and Statistical Tools:** Python, R, SAS, Maple
- **Course Materials and Productivity:** LaTeX, Overleaf, Microsoft Office

TEACHING & STUDENT SUCCESS

- Guided problem solving, active participation, and mathematical discussion
- Formative and summative assessment aligned with course learning objectives
- Use of assessment results to identify and address common misconceptions
- Individualized support for students with different levels of mathematical preparation
- Application-based examples connecting mathematics with scientific and quantitative problems
- Student support through office hours, tutoring, mentoring, and academic guidance
- Technology-supported practice, feedback, and classroom participation
- Critical evaluation of computational mathematical work

ACADEMIC & RESEARCH EXPERIENCE

- Graduate Researcher, University of Alabama at Birmingham** Aug. 2022–Present
 - Conduct research in applied mathematics, mathematical biology, dynamical systems, and longitudinal health-data analysis.
 - Develop and analyze differential-equation models involving infectious and chronic disease.
 - Analyze longitudinal and survival data using Cox proportional hazards and extended Cox models in SAS and R.
 - Prepare manuscripts, technical reports, research presentations, and publication-quality figures.
- Data Analyst Intern, Health Strategy & Delivery Foundation, Nigeria** Jan.–Jul. 2022
 - Analyzed more than 10,000 patient records from the Diabetes Awareness and Care project.
 - Cleaned, validated, and organized health datasets using Python and Microsoft Excel.
 - Identified trends and data-quality issues and prepared quantitative reports for project stakeholders.

RESEARCH PRESENTATIONS

- E. O. Francis**, “Time-Dependent Effects of Leisure-Time Physical Activity on Mortality Among U.S. Adults: A Survey-Weighted Cox Analysis,”
Biostatistics Research Presentation, University of Alabama at Birmingham, Apr. 2026.
- E. O. Francis**, “Optimal Control Analysis Applied to the Transmission Dynamics of Hepatitis B to Primary Liver Cancer,”
Department of Mathematics Seminar, University of Alabama at Birmingham, Mar. 2026.

JOURNAL
PUBLICATIONS

- O. Odetunde, T. Alesinloye, and **O. Francis**, “Modeling the Dynamics of Type 2 Diabetes with Modifiable Lifestyle Influence,” *Journal of the Nigerian Society of Physical Sciences*, 8, Article 3161, 2026.
doi:10.46481/jnsps.2026.3161 .
- O. T. Adedeji, T. S. Alesinloye, and **E. O. Francis**, “A Social Influence-Driven Compartmental Model of Obesity Dynamics with Multi-Class Severity and Relapse,” *Franklin Open*, 16, Article 100698, 2026.
doi:10.1016/j.fraope.2026.100698 .
- O. Odetunde, **O. Francis**, and T. Alesinloye, “Optimal Control Analysis of Hepatitis B Transmission Dynamics,” *World Journal of Advanced Research and Reviews*, 24, 2024.
doi:10.30574/wjarr.2024.24.2.3532 .
- B. Teri, **E. O. Francis**, et al., “The Role of Big Data in Driving Advancements in Deep Learning: Opportunities, Challenges, and Future Directions,” *World Journal of Advanced Research and Reviews*, 24, 690–696, 2024.
doi:10.30574/wjarr.2024.24.2.3390 .

PAPERS IN
PREPARATION

- E. O. Francis**, “Physical Activity, Cardiovascular Disease, and Mortality: A Longitudinal and Mathematical Modeling Approach.”
- E. O. Francis**, “On the Dynamics of HIV–Cancer Comorbidity with Physical Activity-Dependent Mortality.”

SELECTED HONORS

- James R. Ward, Sr. Memorial Endowed Award**, Department of Mathematics, University of Alabama at Birmingham 2026
Awarded for excellence in graduate studies in Differential Equations or Mathematical Biology.
- ICM 2026 Graduate Student Travel Grant**, American Mathematical Society 2026
- Funded Participant, IMSI Opening Tutorial**, *Mathematical and Statistical Foundations of Data Assimilation for Digital Twins* 2025

SERVICE &
OUTREACH

- Volunteer Mathematics Outreach Presenter, “Spill the Mathematical Tea Party,” CAS Open House Week, UAB Department of Mathematics 2026
Led an accessible discussion on *Hilbert’s Hotel and the nature of infinity* for students and visitors.
- Proctor, Alabama Statewide High School Mathematics Contest, University of Alabama at Birmingham 2026
- Proctor, UAB Math Talent Search 2025
- Judge and Grader, UAB Middle School Mathematics Tournament 2024